

Rishabh Shah

New Brunswick, NJ | rs2780@scarletmail.rutgers.edu | (848) 391-0484 | linkedin.com/in/rishabh-shah1 | github.com/RishiiShah | Portfolio

EDUCATION

Rutgers University - New Brunswick

Sep 2025 - May 2027

Master of Science in Computer Science, **GPA: 3.83 / 4.0**

University of Mumbai

Dec 2021 - May 2025

Bachelor of Technology in Artificial Intelligence & Data Science, **Honors: Computational Biology, CGPA: 7.76 / 10.0**

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, Java, TypeScript, SQL, C
- **Frameworks & Tools:** Django, Flask, Next.js, REST APIs, JWT Authentication, Docker, Git, GitHub, GitHub Actions (CI/CD), Linux
- **Databases & Cloud Platforms:** PostgreSQL, MySQL, MongoDB, AWS, Azure, Google Cloud Platform (GCP)
- **AI/ML & Computer Vision:** TensorFlow, PyTorch, Scikit-learn, Keras, Pandas, NumPy, OpenCV, YOLO, OCR, ANPR
- **Coursework:** Data Structures & Algorithms, Operating Systems Design, Intro to AI, Natural Language Processing, Database for Data Science, Massive Data Mining

PROFESSIONAL EXPERIENCE

Aaditya Technologies, Mumbai, India

Jan 2025 – Jul 2025

Software Engineering Intern | Python, Next.js, MySQL, REST APIs, JWT, Docker, AWS, Git, GitHub Actions

- Developed backend services in Python and Next.js API routes integrated with MySQL, optimizing relational database schemas and SQL queries for performance and reliability, reducing average response time by 22%
- Built secure REST APIs with JWT authentication and role-based access control, improving inter-module communication and application security through structured logging
- Containerized 4 services using Docker and deployed to AWS, managing build and deployment workflows with Git and GitHub to maintain environment consistency and reduce deployment friction
- Shipped 10 production-ready features across backend APIs, web application and dashboard modules, writing tests, managing integration and supporting cloud releases on AWS

Creative Line, Mumbai, India

Jun 2024 – Sep 2024

Web Development Intern | Next.js, MySQL, REST APIs

- Refactored 7 reusable components across 3 features, reducing per-file code by nearly 70% and eliminating component coupling
- Developed and tested Next.js API routes with error handling and caching optimizations, improving integration reliability and application performance between frontend and backend services
- Integrated MySQL database with Next.js API routes, designing queries for efficient data retrieval, improving data retrieval speed by 23%, and wrote 2-3 unit tests to validate critical data flows
- Implemented API integrations across the stack following clean architecture and modular design principles, improving long-term maintainability and supporting new feature development

TECHNICAL PROJECTS

Traffic Violation Detection & Automated Ticketing System | IEEE ICICIS 2025

github.com/RishiiShah/Traffic

- Fine-tuned YOLOv12x on a custom dataset through comparative analysis against YOLOv11x to detect 5 traffic violations in real time, achieving 93.76% end-to-end accuracy, 97.56% vehicle type detection and 93.26% helmet detection
- Designed a license plate recognition pipeline using Groq Llama-4 Scout vision model for OCR with Indian number plate regex validation, achieving 92.7% OCR accuracy and 98% localization accuracy
- Automated challan generation with violation metadata, storing evidence frames on AWS S3 and delivering notifications to offenders via Twilio SMS for end-to-end enforcement

Synthetic Financial Time Series Generation via WGAN-GP | Scrivener Publishing 2026

2026

- Designed a WGAN-GP generator with multi-head attention, dual TCN layers, stacked LSTMs and GRU trained on Apple stock data using AdamW over 3000 epochs with log returns and hybrid min-max and standard scaling preprocessing
- Validated synthetic data quality with Pearson and Spearman correlations above 0.80 and R-squared above 0.63 across all price features including Open, High, Low, Close and Adjusted Close
- Augmented real stock data with generated samples, improving downstream prediction model accuracy from 97.93% to 98.83%, demonstrating generalization gains for rare market events like recessions and global crises

JARVIS: Voice Assistant with Smart Home Automation | Journal of Technical Education 2025

github.com/RishiiShah/Jarvis

- Built a 4-module voice assistant on Raspberry Pi 3B+ using Groq hosted Llama 3.1 8B for NLP query processing across deterministic task, math and reminder, LLM query and home automation modules
- Implemented wake word detection, spectral noise filtering, noise gating and multi-threading for concurrent task handling, with prompt engineering to filter harmful queries and protect user privacy
- Engineered the IoT backend using Azure IoT Hub to relay voice commands to Raspberry Pi GPIO pins for real-time appliance control, backed by a Django and Django-Allauth web interface for secure device registration and state management